Wang Ma

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Summary

Ph.D. student working on trustworthy LLMs and agents (uncertainty quantification, abstention, reliable reasoning and RAG), with first-author papers at **NeurIPS 2026** and **CVPR 2026 (Highlight)**. Industry experience improving an LLM agent at Microsoft and developing black-box uncertainty quantification for LLMs at IBM Research. Seeking Summer 2027 research and applied science internships in trustworthy AI, LLM reliability, and generative AI.

Education

Rensselaer Polytechnic Institute (RPI) Aug 2024 – May 2028 (Expected)
Ph.D. in Computer & Systems Engineering (Advisor: Prof. Qiang Ji; also working with Prof. Yao Ma) *Troy, NY*
Research Interests: LLM Uncertainty, Abstention, Reliable Reasoning & RAG, Bayesian Deep Learning, Knowledge-augmented AI

Southern University of Science and Technology (SUSTech) Aug 2020 – July 2024
B.S. in Data Science and Big Data Technology (Advisor: Prof. Chao Wang) *Shenzhen, China*

University of California, Irvine (UCI) Mar 2023 – July 2023
Exchange Student (ASAP Program) *Irvine, CA*

Publications & Manuscripts

1. **Black-Box Uncertainty Quantification for Large Language Models via Ensemble-of-Ensembles**
Wang Ma, Debarun Bhattacharjya, Junkyu Lee, Nhan H Pham, Harsha Kokel, Qiang Ji.
NeurIPS 2026. Preliminary version at AAAI 2026 AIR-FM Workshop.
2. **Towards Knowledge-augmented Bayesian Deep Learning For Computer Vision**
Wang Ma, Hanjing Wang, Yufei Zhang, Darsha Udayanga, Qiang Ji.
CVPR 2026. ***Highlight***.
3. **Distilling Bayesian Uncertainty into a Single Forward Pass**
Wang Ma, Qiang Ji.
Under review at ICLR 2027.
4. **Causal Saliency Map For Post-hoc Model Explanation**
Hanjing Wang, **Wang Ma**, Naiyu Yin, Qiang Ji.
Under review.

Industrial Experience

Microsoft **Redmond, WA**
Applied Science Intern, PPT ML Team *May 2026 – Aug 2026*

- Worked on PowerPoint's **LLM-based slide generation agent**; diagnosed and fixed table-rendering issues in agent-generated slides to improve presentation quality.
- Analyzed agent execution traces to identify wasted iterations in the **build pass**, and implemented a **deterministic runtime tracker** to improve convergence and reduce end-to-end generation latency.

IBM Research **Yorktown Heights, NY**
Research Extern (Mentor: Dr. Debarun Bhattacharjya) *May 2025 – Aug 2025*

- Developed a **perturbation-based UQ framework** for **black-box LLMs**, decomposing uncertainty without access to model weights or intermediate activations.
- Introduced an "ensemble-of-ensembles" method to disentangle **aleatoric and epistemic uncertainty**, with AUROC comparable to or better than white-box baselines; first-authored paper accepted to **NeurIPS 2026**.

Academic Experience

Rensselaer Polytechnic Institute (RPI) **Troy, NY**
Graduate Research Assistant (Advisor: Prof. Qiang Ji) *Sept 2024 – Present*

- **Knowledge-Augmented Bayesian Deep Learning:** Proposed a two-stage framework with a **knowledge-induced prior** and **adaptive knowledge likelihood**, improving accuracy, robustness, and **OOD detection**.
- **Single-Pass Bayesian Uncertainty:** Distilled Bayesian ensemble uncertainty into a single-forward-pass student via a two-stage knowledge distillation pipeline, enabling efficient uncertainty estimation at inference time.
- **Causal Saliency Maps:** Implemented **Causal Markov Blanket (CMB)** detection algorithms using label-wise mutual information to identify causally relevant features and filter spurious correlations.
- **Semantic Uncertainty Disentanglement:** Developed a **contrastive learning framework** to disentangle semantic and nuisance uncertainties, achieving strong benchmark performance. (Technical Report)
- **Efficient Single-Model UQ:** Used **NTK theory** and **credal interval-based methods** to study practical uncertainty quantification with a single auxiliary neural network. (Technical Report)

SUSTech **Shenzhen, China**
Undergraduate Researcher (Advisor: Prof. Chao Wang) *Mar 2024 – Aug 2024*

- **Hyperspectral Image Restoration:** Built a self-supervised method combining **Implicit Neural Representations (INR)** and **Diffusion Models**.
- **Unpaired Image Denoising:** Optimized a **VAE** from unpaired data with Mutual Information maximization.

Technical Skills

- **Programming:** Python (PyTorch, NumPy, Pandas, Scikit-learn), Java, MATLAB, LaTeX
- **Developer Tools:** Git, Docker, Linux
- **Natural Languages:** Mandarin Chinese (Native), English (Fluent), Japanese (Conversational)

Honors & Service

- **Golden Reviewer**, International Conference on Machine Learning (ICML) 2026
- **SUSTech Excellent Undergraduate Graduation Project** (Thesis: End-to-end Unpaired Image Denoising) June 2024
- **Seminar Organizer:** AI Optimization & Theory (SUSTech); led sessions on Bayesian Optimization & Bayesian Deep Learning 2024
- Second Prize in Guangdong Province, Mathematical Contest in Modeling 2021
- **Scholarships:** SUSTech Excellent Student Scholarship (2021, 2022)